Biomass and Bioenergy 204 (2026) 108476

Contents lists available at ScienceDirect

Biomass and Bioenergy

journal homepage: www.elsevier.com/locate/biombioe

Transforming *Agave sisalana* waste into high-yield biogas: An approach to sustainable energy

Fabiane M. Vieira [a,b], Maiki S. de Paula [d], Maria Paula C. Volpi [c], Oscar F. Herrera Adarme [b], Thiago Ribas [d], Jean C.G. Silva [b], Marcelo F. Carazzolle [d], Gonçalo A.G. Pereira [d,*,1], Gustavo Mockaitis [b,**,1]

[a] *Interinstitutional Graduate Program in Bioenergy (USP/UNICAMP/UNESP), 330 Cora Coralina Street, Cidade Universitária, Campinas, SP, CEP 13.083-896, Brazil*

[b] *Interdisciplinary Research Group on Biotechnology Applied to the Agriculture and the Environment, School of Agricultural Engineering, University of Campinas (GBMA/FEAGRI/UNICAMP), Campinas, Brazil*

[c] *Department of Biosystems Engineering, LEB, Luiz de Queiroz College of Agriculture - Esalq, University of São Paulo - USP, Av. Pádua Dias, 11 - Agronomy, Piracicaba, SP, 13418-900, Brazil*

[d] *Department of Genetics and Evolution, Microbiology, and Immunology, Laboratory of Genomics and bioEnergy (LGE), Institute of Biology (IB), University of Campinas (UNICAMP), Campinas, São Paulo, Brazil*

ARTICLE INFO

Keywords:
Methane
Agave residues
Sisal pulp
Sisal juice
Byproducts
Agave sisalana

ABSTRACT

Agave, a native Mexican plant widely used for fiber extraction and alcoholic beverage production, has significant potential for biogas generation. Brazil, the global leader in sisal fiber production, extracts fiber from only 4 % of the agave leaf, leaving approximately 24 tons of organic residue per ton of fiber. These residues can be converted into methane (CH_4) through anaerobic digestion (AD). This study explores biogas production from agave fiber extraction residues. The juice and pulp contain high levels of calcium, potassium, magnesium (2–3 gL^{-1}), citric acid (56 $g\ L^{-1}$), and formic acid (76 $g\ L^{-1}$). Methane potentials were 476 $NmLCH_4\ gVS^{-1}$ for the juice and 331.10 $NmLCH_4\ gVS^{-1}$ for the pulp. Anaerobic digestion revealed microbial groups such as *Anaerolineae*, *Synergistia*, *Clostridia*, *Bacteroidia*, and *Gammaproteobacteria*, which degrade carbohydrates and volatile fatty acids. *Halobacteriota*, a methane-producing archaea, predominated in all samples. Energy recovery scenarios considering raw sisal pulp direct AD (1) and an integrated biorefinery proposal (2) suggests that 1034 MJ of energy can be obtained per ton of biomass, reaching 1951 MJ when integrated into a biorefinery. This study emphasizes the value of agave defibration residues, typically discarded, and proposes an AD-based biorefinery model to enhance the sustainability of sisal fiber production.

1. Introduction

Recent global advancements address greenhouse gas emissions towards renewable energy generation. The Paris Agreement, ratified at COP 21, focusses in achieving net-zero carbon emissions by 2050. Thus, there has been a pursuit for alternative bioenergy sources. Biogas emerged as a clean energy source, considering biomethane production and electricity generation [1]. Consequently, biogas holds a significant promise as a resource for decarbonization efforts.

Biogas is a methane (CH_4)-rich gas with substantial energy potential (calorific value of 35 $kJ\ Nm^{-3}$). Biogas is produced through anaerobic digestion (AD) of complex organic matter, widely present in urban, agriculture and industrial residues and in biomass. In Brazil, due to high activity in the agricultural sector, main feedstocks for AD comprises byproducts or specific residues, such as: vinasse, municipal solid waste, sewage, and animal manure, among other types of plant sources [2–4]. Most of these residues originate from Brazil's central-southern region, where agricultural activities are more intensive, leading to higher levels of industrial-scale biogas production compared to other regions [5].

Among plant biomasses of interest for AD, Agave is an emerging topic in literature. Native to Mexico, Agave has gained attention as a research subject for AD towards biogas production from its by-products

* Corresponding author.
** Corresponding author.
E-mail addresses: goncalo@unicamp.br (G.A.G. Pereira), gusmock@unicamp.br (G. Mockaitis).
[1] Shared Senior Authorship.

https://doi.org/10.1016/j.biombioe.2025.108476
Received 14 July 2025; Received in revised form 8 October 2025; Accepted 11 October 2025
Available online 15 October 2025
0961-9534/

[6]. Due to its low water requirements, this plant has thrived and shown robust development in the Brazilian Northeastern semiarid regions.

In Mexico, Agave's heart (piña) is primarily used to produce alcoholic beverages, such as tequila, mezcal, bacanora, raicilla, sotol, and pulque, and the rest of the plant is discarded as a residue. However, agave now has various alternative applications, utilizing the entire plant. Agave can be used as an effective living barrier, enhancing soil pH, elevating phosphorus, and potassium concentrations, and preventing erosion. Conversely, diverse by-products offer a multitude of uses [7].

Leaves, bagasse, and fibers represent the principal consequential by-products derived from the agave plant. Leaves, characterized by lignocellulosic composition, consist predominantly of robust and coarse fibers, traditionally employed in textile and fabric production. Particularly, varieties like *Agave sisalana* (sisal), *Agave mapisaga*, and *Agave salmiana* yield resilient fibers highly valued for their durability in ethnic clothing and cordage [7].

Brazil is a leading producer of agave plants, primarily cultivated for fiber extraction. According to FAO (2024), Brazil is the largest producer of sisal fiber, holding 33.5 % of global production. Sisal fiber production in Brazil is based on *Agave sisalana* and occurs mainly on small farms using a decentralized and mobile dry decortication method. This process generates a substantial production of organic residues, with poor waste management. Since only 4 % of sisal leaves comprises fiber, each ton of fiber produces approximately 18–24 tons of organic residues, which are often left in fields or utilized as animal feed. These practices may result in diverse environmental impacts, particularly when the residue is left in fields. These environmental impacts are not completely understood for the production of sisal fiber in Brazil. Based on Brazil's 2023 sisal fiber production (95,567 tonnes), and the widely reported material balance that only 4 % of leaf mass becomes fiber, approximately 24 tonnes of leaf residues are generated per 1 ton of fiber production, leading to a generation of 2.29 million t of sisal leaf residues in 2023. Sisal pulp (SP), a waste of agave fiber extraction, consists of leaf tissues with small amounts of discarded sisal fibers. Although this byproduct could serve as a valuable food supplement for animals, this residue remains unused and left in the soil in the properties. If this material is left on the ground or flushed to unlined ditches, the decortication liquid and pulp can deplete oxygen in receiving waters, pollute surface and groundwater, and emit methane during anaerobic decay; several assessments and LCAs flag these impacts and recommend valorization pathways [5].

Due to its high organic matter content, this neglected residue can cause negative effects on the soil, but also presents significant potential for biogas production [8]. Colley et al. [9] demonstrated that AD of *A. sisalana* pulp produced CH_4 at rates ranging from 0.18 to 0.48 m^3 CH_4 kg volatile solids^{-1} (VS). Soares et al. [24] reported CH_4 yields of 113.9 NmL CH_4 g O_2^{-1} from *A. sisalana* juice. Arreola-Vargas et al. [6] investigated AD of *A. tequilana* pineapple bagasse (residue from the tequila industry) using a bench-scale bioreactor, yielding 0.26 L CH_4 g O_2^{-1}. These studies highlight the potential for energy recovery from agave byproducts, particularly methane from bagasse and pulp residues, which corresponds to 96 % of *A. sisalana* leaves after fiber extraction.

This study proposes an integrated and sustainable biorefinery model that addresses the increasing demand for solutions to agro-industrial waste management. The separation of liquid and solid fractions from *Agave sisalana* by-products enhances biomass utilization by enabling biogas and biochar production, thereby minimizing waste and fostering a circular economy.

Given the promising results and the importance of agave culture towards establishing biorefineries in the northeastern region of Brazil, this paper reports CH_4 production experiments from fiber extraction residues of *A. sisalana*. CH_4 production was assessed for *A. sisalana* pulp and *A. sisalana* juice, as solid and liquid by-products from decortication, respectively. The study compared methane yields and energy recovery potential between raw SP and sisal juice (SJ) while also analyzing the microbial community and its relationship with metabolic pathways of AD.

2. Materials and methods

2.1. Inoculum and substrates

The inoculum used in this study was sourced from an Upflow Anaerobic Sludge Blanket (UASB) bioreactor at the Ideal poultry slaughterhouse in Pereiras, São Paulo state, Brazil (SISGEN: AFC1691).

Agave sisalana leaf by-products (pulp and juice) were obtained through the decortication process (Fig. 1). Agave plants were obtained from Conceição do Coité, Bahia state, Brazil. The leaves were processed using a defibrating machine to extract the leaf fibers (sisal fiber), separating them from the pulp. Raw SP is constituted of SJ and dry SP as mixed solid parts. SJ was obtained from raw SP by mechanical pressing without any addition of water, separating it from the dry SP. Both raw SP – without pre-treatments or drying steps – and SJ were processed as byproducts used for CH_4 production assessment.

2.2. CH_4 batch assays

The investigation of CH_4 production was carried out in 325 mL Schott® Duran flasks, with a working volume of 150 mL. Flasks were tightly sealed with rubber stoppers. All experiments were conducted in shaker incubators with temperature control, at mesophilic conditions (35 °C), with orbital continuous stirring (150 rpm). The experiments used an inoculation ratio of 2:1 (substrate:inoculum), in terms of volatile solids (VS). Macro and micronutrients [10] were added to each flask ensuring optimal conditions for microbial process. The flasks with SP were prepared targeting a final 5 % concentration of the total solids (TS) and the SJ flasks was diluted 10x to carry out the experiments [11]. After assembling, N_2 gas was fluxed through each flask for 1 min, creating an inert headspace and ensuring anaerobic conditions. All experiments were performed in triplicate, with the initial pH set at 7. No active pH control was applied during the batch assays; pH was recorded at the start and at the end of the tests to confirm that acidification did not occur. Endogenic CH_4 production was estimated through a negative control without substrate addition.

Pressure was measured using a differential pressure manometer (MAN30, Amprobe). Biogas component concentrations was determined using a Shimadzu Nexis™ GC-2030 gas chromatograph. N_2, and He were used as carrier gases, at pressures of 180.0 kPa and 310.0 kPa, respectively. Chromatographic columns were maintained at 60 °C, with detectors 1 (for CH_4, H_2S, and CO_2 measurement) and 2 (for H_2 measurement) operating at 150 °C.

The CH_4 batch experiments were finished when the production was less than 5 % of the cumulative production [12]. All biogas production was expressed at standard temperature and pressure (STP) conditions (273.15 K and 101,325 Pa).

2.3. Physicochemical analysis

Physicochemical analyses were performed in the substrates and inoculum, accordingly 'Standard Methods for the Examination of Water and Wastewater' [13]: total solids (ST – 2540B), volatile solids (VS – 2540E), chemical oxygen demand (COD – 5220B) and pH (4500B). Total organic carbon (TOC) and total nitrogen (TN) were determined through ASTM D7573-18a, using a TOC Shimadzu® equipment with a coupled TN module.

2.3.1. Liquid chromatography

Volatile fatty acids (VFA), alcohols, and carbohydrates were determined for liquid phase samples and digestates employing high-performance liquid chromatography (HPLC). Samples were filtered through a 0.22 μm PVDF syringe filter and acidified with 40 μL of H_2SO_4 1N. The HPLC system (Shimadzu) comprised a pump module (LC-

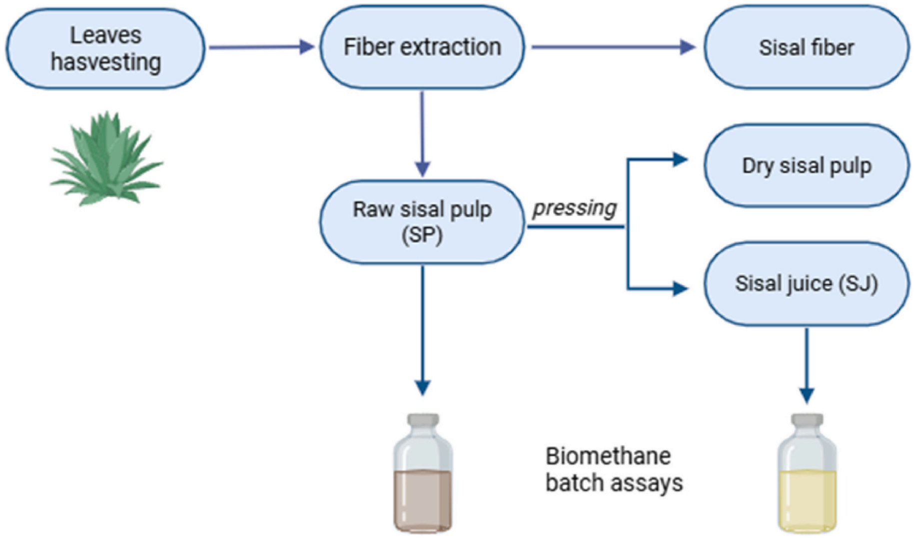

Fig. 1. Diagram of *A. sisalana* residues used for AD in this study.

10ADVP), an automatic sampler (SIL-20A HT), a column oven (CTO-20A) set at 43 °C, a diode array detector (SDP-M10 AVP), and an Aminex HPX-87H column (300 mm length, 7.8 mm inner diameter, Bio-Rad). Detection utilized a refractive index detector. The mobile phase was 0.01 N H_2SO_4, with a flow rate of 0.5 ml min^{-1}.

2.3.2. Macro and microelements analysis

For the determination of macro and microelements, ion detection analyses were conducted, involving the assessment of ion concentrations. Elemental analysis was performed using Inductively Coupled Plasma-Optical Emission Spectrometry (ICP-OES) and elemental analysis techniques.

For the analysis of chemical elements agave juice and pulp and inoculum samples without pre-treatment were used. The United States Environmental Protection Agency (USEPA) Method 3015A was applied, wherein 5 mL of subsamples underwent digestion in a microwave oven with a strong acid mixture (HNO_3 and H_2O_2) or a 20 min treatment at 170 °C, following USEPA guidelines (USEPA, 1986). The comprehensive metal composition was subsequently determined using inductively coupled plasma-atomic emission spectrometry (ICP-AES) on a Thermo Fisher Scientific, Waltham, MA, USA, as detailed by Hosseini et al. (2019) [14].

2.4. Kinetics

For kinetic adjustment, the modified Boltzmann and the Gompertz modified model were employed, a well-established approach utilized in prior research conducted by our research group [4]. The adjustment was performed to assess the temporal evolution of CH_4 volumetric production. Equations (1) and (2) encapsulate the representation of the employed models.

$$V_{CH4}^{STP} = \frac{V_{CH4}^{max}}{1 + e^{\left(\frac{4r_m*(t1-t)}{V_{CH4}^{max}}\right)}} \qquad \text{Equation 1}$$

$$V_{CH4}^{STP} = V_{CH4}^{max}\, exp\left\{-exp\left[\frac{r_m \cdot e}{V_{CH4}^{max}} \cdot (\lambda - 1) + 1\right]\right\} \qquad \text{Equation 2}$$

V_{CH4}^{STP} denotes the specific CH_4 production over time (in NmL CH_4 g VS^{-1}), V_{CH4}^{max} represents the maximum specific volumetric production achieved throughout the experiment (in NmL CH_4 g VS^{-1}). The term 't_1' denotes the time point at which the peak rate of production is reached for the first sigmoidal pattern (measured in days), and 'r_m' indicates the maximum rate of CH_4 production during this initial sigmoidal pattern (measured in NmL CH_4 g VS^{-1} day^{-1}). The parameter 'λ' (days) denotes the lag phase duration. Analysis of variance (ANOVA) and Tukey's test were performed at a 95 % confidence level on the experimental methane production values to evaluate whether significant differences existed between the treatments.

2.5. Energy analysis for agave biorefinery flowchart using experimental data

To evaluate the energy recovery in the context of A. *sisalana* biorefinery using AD experimental data obtained, an energy analysis was performed. To better represent a biorefinery system, it was considered that raw SP is divided into SJ and dry SP (Fig. 1). Thus, two primary scenarios were identified in this study: Scenario 1, biomethane ($bioCH_4$) is produced directly from AD of the SP. Scenario 2, SJ and dry SP are separated, with the dry SP been included in the pyrolysis process according with typical operational conditions for obtain biochar [15] and SJ been used for biomethane production. Both scenarios calculate the maximum potential energy generation from 1 ton of harvested leaves, considering that 96 % of the leaves are SP, which consists of 80 % SJ and 20 % dry SP. To evaluate the energy recovery from CH_4 production during AD of agave juice and pulp, the calculations were conducted considering the energy generated using biogas in a Combined Heat and Power system (CHP). The energetic demands considered were the employed in the fiber extraction process and biogas purification along with all the other unit operations [16]. The energy production from AD (E_P^{AD}) for each ton of leaf processed was calculated according to Equation (3).

$$E_P^{AD} = Max_{CH_4} \,.\, VS_{substrate} \,.\, 34.5\ MJ\ Nm^3 CH_4^{-1} \qquad \text{Equation 3}$$

Where E_P^{AD} is the energy production in MJ; $VS_{substrate}$ is the volatile solids of SP or SJ (kgVS); Max_{CH_4} is maximum experimental potential of CH_4 production (Nm^3 CH_4 $kgVS^{-1}$) and 34.5 MJ $Nm^3CH_4^{-1}$ is lower calorific value (LCV) of biomethane ($bioCH_4$). On the other hand, the energy obtained from pyrolysis process was calculated using Equation (4), where $Q_{pyrolytic\ gas}$ is the pyrolytic gas flow obtained during dry SP processing, and 9.2 MJ kg^{-1} is the calorific value of the pyrolytic gas at 500 °C [17].

$$E_P^{Py} = Q_{pyrolitic\ gas} \,.\, 9.2\ MJ\ kg^{-1} \qquad \text{Equation 4}$$

The typical efficiencies of a CHP system were considered for the conversion of E_P^{AD} to electric (E_E) and thermal (E_T) energies, as shown in Equation (5) and Equation (6).

$$E_E = E_P^{AD} \times 0.85 \times 0.33 \quad \text{Equation 5}$$

$$E_T = E_P^{AD} \times 0.85 \times 0.66 \quad \text{Equation 6}$$

Where 0.85 is the regular efficiency of the combustion engine; 0.33 is the conversion factor of E_P^{AD} to E_E and 0.66 is the conversion factor of E_P^{AD} to E_T [18]. The energy consumption for biogas purification was estimated at 10 % of total electricity produced.

2.6. DNA extraction and 16S rRNA gene amplicon sequencing

Molecular biology analysis was conducted on the microbial community of seed sludge and each flask at the end of the experiments. For DNA extraction, samples underwent homogenization by vortexing, with 800 g of solid samples and 600 μl of liquid samples used for triplicate extractions. The DNeasy PowerSoil Pro kit (QIAGEN) was utilized following the manufacturer's protocol. Quantification of DNA samples was carried out using a Qubit® 3.0 Fluorometer (Life Technologies), while quality assessment based on the 260/280 ratio was determined using the NanoDrop Lite (Thermo Scientific). V3 and V4 regions of 16S rRNA gene were amplified with primers 341f (5′-CCTACGGGNGGCWGCAG-3′) and 785r (5′-GACTACHVGGGTATCTAATCC-3) and the sequencing was conducted using paired-end reads (2x300bp) on the Illumina Miseq.

Amplicons sequences were submitted to SRA database under project PRJNA1147732 using the following accession numbers: sj_1 (SRR30224875), sj_2 (SRR30224874), sj_3 (SRR30224873), sp_1 (SRR30224872), sp_2 (SRR30224871), sp_3 (SRR30224870), neg_control (SRR30224869), seed_sludge (SRR30224868)

2.7. Metataxonomic analysis

Quality control of raw sequencing reads was performed using Fastp (version 0.23.4) under default settings, removing reads shorter than 50 base pairs [19]. Raw reads were then demultiplexed using QIIME 2 (version 2024.2.0), and further processed with the q2-dada2 plugin for denoising and chimera removal, with truncation set at position 293 on the reverse reads [20,21].

For taxonomic classification and diversity analysis: 1) the SILVA ribosomal database version 138-99 [22] was used to train a Naive Bayes classifier, which assigned taxonomy to the denoised sequences; 2) Feature tables (ASV tables) and taxa bar plots were generated, including the filtration of mitochondrial and chloroplast contaminants, and rarefaction was applied to account for differences in sequencing depth; and 3) Alpha diversity was assessed using Shannon diversity, while beta diversity was analyzed through Principal Coordinate Analysis using Bray-Curtis distance, all visualized with QIIME 2 [23,24].

3. Results and discussion

3.1. Characterization of residues and inoculum

Table 1 presents the main results of physicochemical analysis for the substrates and inoculum. The COD of SJ was 80.33 g O_2 L^{-1} which means a high content of organic matter to be consumed and converted into volatile fatty acids and biogas, in comparation to vinasse, by-product from sugarcane ethanol production and is commonly used to biogas production in Brazil, has a COD from 20 to 30 g O_2 L^{-1} [25]. Furthermore, this value aligns with the findings of Soares et al. (2023) [26] which yielded a COD of 96.05 g O_2 L^{-1}.

It is also possible to notice that there is a high content of sugars and organic acids, which can also be used for biogas production. A large amount of glucose (16 g) holds great potential for biogas generation, as it will be broken down in hydrolysis and subsequently utilized for organic acids, potentially generating acetic acid, which is the main precursor of CH_4. In the study conducted by Soares et al. (2023) [26], report carbohydrates of 14.6 g L^{-1}, which denotes the sum of soluble mono-/disaccharides and short-chain oligosaccharides as total carbohydrate. Thus, glucose is different from total carbohydrates and the two numbers are not strictly interchangeable; nevertheless, both metrics consistently indicate a sugar-rich feedstock compatible with high methane potential.

Another important organic compound for AD is methanol, which is present at a concentration of 10 g L^{-1} in SP. Recent discoveries have identified methanogens capable of reducing methanol to CH_4 using hydrogen [27]. Notably, classes *Thermoplasmata* and *Methanobacteria*, both capable of utilizing this pathway, were identified in the metataxonomic analysis of this study (section 3.4).

The C/N ratio of SJ was 48, which is a little higher than the ideal range of 20–35 for optimal AD performance [28]. To address this imbalance, a micronutrient solution was added [10], since the nitrogen concentration in SJ was extremely low, resulting in poor nutritional balance. After adding the nutrient solution, the C/N ratio for the SJ experiments improved to approximately 15, bringing it closer to the ideal range. Similarly, the SP experiment also required the addition of a micronutrient solution due to its high initial C/N ratio of 276. After nutrient addition, the C/N ratio was of 12, which is closer to the ideal range, although still slightly lower than optimal. After the nutrient solution addition, the C/N of both substrates remained outside the optimal range for anaerobic digestion, because the solution used here primarily supplies trace elements/alkalinity rather than nitrogen, and its N dose was intentionally kept low to avoid ammonia accumulation in BMP tests.

Table 1
Physicochemical characterization of sisal residues and inoculum.

Parameters	SP	SJ	Inoculum
TOC	0.407 g g_{sp}^{-1}	22.93 g L^{-1}	NA
TN (g L^{-1})	–	0.48	NA
Total Solids[a] (g kg^{-1})	151.42 ± 2.94	62.23 ± 0.04	59.70 ± 3.01
Volatile Solids[a] (g kg^{-1})	98.39 ± 3.05	47.93 ± 0.04	50.23 ± 2.59
COD[a] (g O_2 L^{-1})	ND	80.33 ± 1.93	NA
Citric acid (g L^{-1})	NA	56.09 ± 2.99	NA
Malic acid (g L^{-1})	NA	8.47 ± 0.42	NA
Butanedioic acid (g L^{-1})	NA	1.10 ± 0.05	NA
Formic acid (g L^{-1})	NA	76.81 ± 4.05	NA
Glucose (mg L^{-1})	NA	16.19 ± 0.81	NA
Fructose (mg L^{-1})	NA	9.86 ± 0.45	NA
Methanol (mg L^{-1})	NA	10.93 ± 0.08	NA

[a] Mean of three replicates ± standard deviation; ND: not determined; NA: not applicable.

Table 2
Microelement characterization of SJ and SP.

Element	SJ (mg L^{-1})	SP (mg kg^{-1})
Aluminum (Al)	0.76	22,513.50
Boron (B)	8.38	5.90
Barium (Ba)	9.98	48.90
Calcium (Ca)	2357.33	20,337.80
Cobalt (Co)	0.02	1.20
Copper (Cu)	0.54	96.20
Iron (Fe)	12.36	11,828.40
Potassium (K)	2368.00	2256.80
Magnesium (Mg)	2714.67	3248.60
Manganese (Mn)	2.49	557.30
Molybdenum (Mo)	0.04	0.90
Sodium (Na)	58.71	606.60
Phosphorus (P)	416.09	9171.60
Sulfur (S)	141.60	2652.70
Silicon (Si)	0.94	834.10
Titanium (Ti)	0.15	153.00
Zinc (Zn)	3.61	143.90

Table 2 presents the inorganic elements in the substrates that can be considered micronutrients for AD. It is noticeable that prevailing inorganic elements are Aluminum (Al), Calcium (Ca) and Iron (Fe) for SP and Ca, Magnesium (Mg) and Potassium (K) for SJ. Literature reports that Fe is responsible for the synthesis and activation of a wide variety of enzymes involved in AD, as for example ferredoxins [29], crucial for the degradation of complex organic matter and is predominant in methanogens, it plays a pivotal role in CH_4 production. Even though it is important to different metabolic pathways, Fe is also reported to be a problem in concentrations higher than 1.3 g L^{-1} [30], which can be a problem for SP.

Excessive Al concentrations can also inhibit anaerobes, as showed by Berzal de Frutos et al. (2023) [31] in which reported that low Al concentrations may not affect AD, but high concentrations such as 306 mg L^{-1} may corroborate with a VFA accumulation, and reducing biogas production. In SP, this concentration is also much higher than indicated by the literature, potentially leading to inhibition issues in CH_4 production. As it concerns the Ca concentration, Ahn et al. (2006) [32] reported that concentrations from 0 to 3 g L^{-1} of Ca had a positive impact on the duration of lag phase, presenting a higher CH_4 concentration when Ca was at 3 g L^{-1} which may have corroborated with the absence of lag phase in this study. Mg is important for the activation of kinases and phosphotransferases, and even in higher concentrations, it exerts minimal influence on CH_4 production, resulting in only a slight reduction in biogas production within the range of 1–20 g L^{-1}, while promoting CH_4 production at higher concentrations. For both substrates (SP and SJ), the lack of Mg is evident, underscoring the importance of adding micronutrients to the AD process [33]. K is a key metal linked to cellular growth and plays a pivotal role in ensuring biological functions align with expectations. Literature suggests an ideal concentration not exceeding 400 mg L^{-1}, as higher concentrations are reported to be detrimental to microbial activity. The concentrations of K exceed the ideal levels reported in the literature for both substrates. However, it is important to understand that micronutrients need to be balanced, and that slightly higher concentrations than recommended may not necessarily be inhibitory which is different from when continuous or semi-continuous systems are evaluated. The soluble K^+ originating from SP can leach into the liquid phase and accumulate via recycle streams, potentially approaching inhibitory levels [34].

Another element that is important in AD is S that appears in high concentration in SP (2 g L^{-1}), which may lead to the formation of hydrogen sulfide (H_2S). Sulfate-reducing bacteria (SRBs) convert sulfate to H_2S in anaerobic environments, posing a problem in anaerobic digestion. The main practical implications of sulfidogenesis include losses in the energy potential of biogas and inhibiting conditions by H_2S for anaerobic microorganisms, including methanogens, disrupting methanogenesis. In AD systems, SRBs and methanogens compete for hydrogen and acetate. SRBs usually outcompete methanogens due to their faster growth rate and higher substrate affinity. Additionally, converting hydrogen and acetate to H_2S is more thermodynamically favorable than converting them to CH_4, making sulfidogenesis more dominant in sulfur-rich environments [35].

In addition to trace elements present in large quantities, it is evident that other nutrients such as Co, Cu, Mo, and Zn are present in very low concentrations, however, even these small concentrations found in this study fall within the ideal ranges considered for AD, indicating that they would not negatively affect the development of the process [36]. An essential trace element that was not detected in this characterization is Nickel (Ni) that is a component of the coenzyme F_{430}, important in the CH_4 formation in all methanogenesis pathways [36]. The scarcity of these micronutrients highlights the need for nutritional supplementation in these substrates to ensure optimal process performance.

3.2. CH_4 production using SJ and SP

Fig. 2 presents the CH_4 production obtained from SP and SJ, respectively. In this study, the maximum CH_4 production was 331.10 NmL gVS^{-1} for SP and 476.00 NmL gVS^{-1} for SJ. Even though the CH_4 production for SJ was higher than for SP, which is expected due to the best performance of mass transfer processes in liquid AD, the values were not significantly different at 5 % probability at Tukey. It can also be observed that the experimental error is higher for the SP triplicates compared to the SJ triplicates, likely due to the heterogeneity of the substrate.

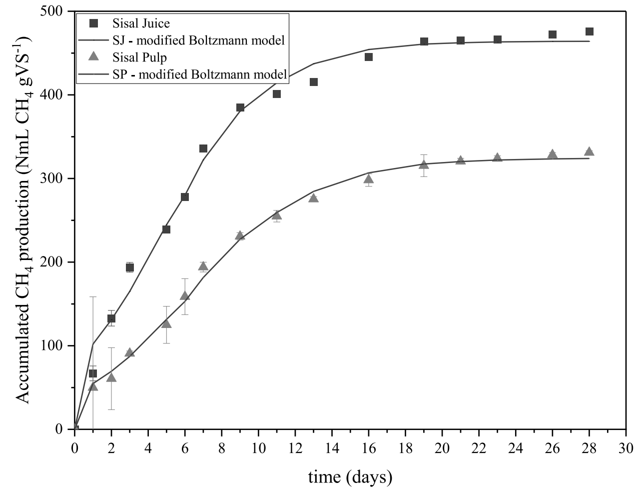

Fig. 2. CH_4 production from *Agave sisalana* juice and pulp (SP) and Boltzmann model adjusted.

Cundr and Haladova (2015) [37] found that when co-digesting SP and zebu dung the CH_4 production reached 173 mL CH_4 gVS^{-1}, which is approximately a half the CH_4 yield obtained in this study using SP. Literature regarding the utilization of Agave sp. residues, particularly those obtained from leaves, is limited; however, a few studies have documented CH_4 production from crassulacean metabolism (CAM) plants [38], which are plants with metabolism similar to that of agave, achieving a CH_4 production ranging from 280.5 to 381.7 NmL CH_4 gVS^{-1}. Notably, the highest yield, 381.7 NmL CH_4 g VS^{-1}, was achieved using crushed leaves of *Agave angustifolia* with a lag phase of 18 days [38]. Other studies conducted by Rizzioli et al. (2024) [39] using minced cladodes of *Opuntia ficus-indica*, another CAM plant, reported production of 210 mL CH_4 gVS^{-1} in semi-continuous AD after 20 days.

In the study by Soares et al. (2023) [26], AD of diluted SJ was conducted in two stages to produce CH_4 and hydrogen, as well as in a single-stage batch process without the addition of micronutrients. They reported direct CH_4 production (similar to this study) as 113.9 NmL CH_4 per gram of O_2. Converting the CH_4 value from this study to grams of COD results in a production of 317 NmL CH_4 per gram of O_2, which is 80.5 % higher than the value reported. The difference in these values is due to the addition of micronutrients in this study, which were not used in the aforementioned study. As detailed in Section 3.1, agave juice has nutritional deficiencies for AD.

Comparing CH_4 production from agave with a residue from an energy crop commonly used in Brazil, which is sugarcane vinasse, CH_4 production is quite similar to SJ. In the study reported by Volpi et al. (2021) [4], the CH_4 production of vinasse was 476 NmLCH_4 gVS^{-1}, while that of SP is comparable to the CH_4 production obtained for filter cake, which was 353 NmLCH_4 gVS^{-1}, considering they are solid residues and share the same mass transfer difficulties and limitations.

Another solid residue commonly used for CH_4 recovery is agave bagasse from tequila production. Typically, converting these sugars into CH_4 requires pretreatment strategies. For instance, ruminal fluid pretreatment in agave bagasse resulted in CH_4 production values of 78 ± 2 mL CH_4 gTS^{-1} [40]. Chemical and biological pretreatments in this same residue yielded a value of 219 mL CH_4 g O_2^{-1} [41]. Untreated lignocellulosic biomass from agave bagasse shows a production of 128 ± 4 mL CH_4 g O_2^{-1}. This indicates that using SJ is more efficient, as it does not require pretreatment steps and results in higher CH_4 recovery. In light of this, there is no need to process SP or using biomass pretreatments which can affect operational and capital expenditure in biorefineries [42].

Table 3 shows the kinetic parameters for the AD conducted in this study. A shorter t_1 presented when comparing SJ and SP represents a smaller adaptive phase and higher daily production from this residue, which is expected when comparing liquid with solid or semi-solid substrates. In study from Soares et al. (2023) [26] that carried a CH_4 production from *A. sisalana* juice (113.9 NmL CH_4 g O_2^{-1}), the adaptive phase was 9.5 days, while this study presents a higher CH_4 production with no lag phase, which is indicated by the parameter λ in the Gompertz model.

Even though this study used two different models to perform the kinetics, the modified Boltzmann model was the one with better fitting of the experimental data. The better fitting is indicated by the lower AIC and RMSE and higher R^2 presented in Table 3. One of the most used parameters for the better fitting of the model is V_{max} and r_m were higher for SJ in the Boltzmann model, confirming the influence of solids content in CH_4 production rates.

Fig. 3 shows the organic acids and sugars identified at the end of the anaerobic batch experiments. Citric, malic, and butanedioic acids were present in high quantities in the initial characterization of SJ (Table 1) and remain in significant quantities at the end of the experiment. This indicates that these acids are naturally abundant in the substrate. Although they were partially consumed during the experiment, they remained in traceable amounts. Sugars were not identified at the end of SJ experiments, indicating that the extent of the experiment was enough to consume the sugar content. The remaining glucose content at the end of SP experiments may be explained due to the longer time to degrade cellulosic matter, or complex sugars that are commonly identified in agave species [43].

The presence of various organic acids (Fig. 3) indicates that a propionic pathway was likely one route for CH_4 conversion, evidenced by the high concentration of formic acid in SJ [44]. Another potential pathway for propionic acid generation is the decarboxylation of succinate, which is also present in elevated quantities in SJ [45]. During the initial phases of AD, these organic acids were converted into acetate, H_2, and CO_2 by acetogenic bacteria, which then led to CH_4 production by methanogens.

3.3. Metataxonomic profile of agave's anaerobic digestion

The microbial communities from the initial seed sludge, negative control (inoculum without substrate), SJ (sisal juice), and SP (sisal pulp) digestion assays were investigated to identify the taxa potentially involved in the anaerobic digestion of *A. sisalana*. A total of 986 ASVs were identified in the entire dataset, 922 from Bacteria and 64 from Archaea. The relative abundance at the class level of each sample is presented in Fig. 4.

Table 3
Kinetic model parameters of A. *sisalana* residues.

Model	Parameters	*A. sisalana* Juice	*A. sisalana* Pulp
Modified Boltzmann	V_{max} (NmL CH_4 g_{VS}^{-1})	464.25	324.41
	t_1 (days)	4.6	6.3
	r_m (NmL CH_4 g_{VS}^{-1} day^{-1})	39.18	24.07
	RMSE	14.28	6.22
	NRMSE	3.00	1.88
	AIC	91.09	64.48
	R^2	0.991	0.996
	R^2_{adj}	0.988	0.994
Gompertz	V_{max} (NmL CH_4 g_{VS}^{-1})	463.23	328.82
	λ (day)	0	0
	r_m (NmL CH_4 g_{VS}^{-1} day^{-1})	50.94	26.67
	RMSE	19.02	8.07
	NRMSE	3.99	2.44
	AIC	100.25	72.83
	R^2	0.985	0.994
	R^2_{adj}	0.979	0.992

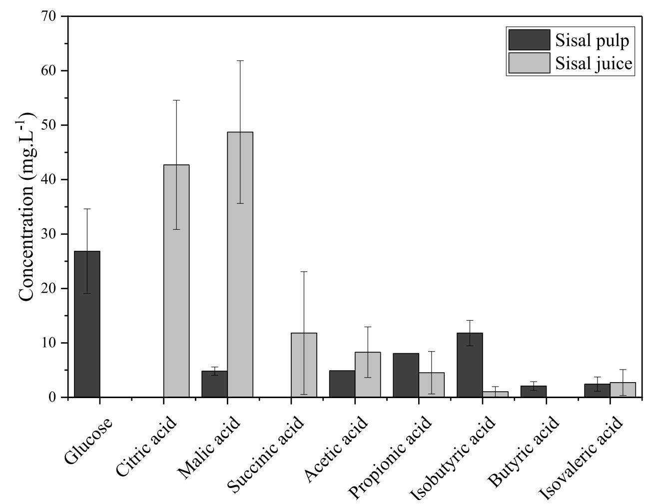

Fig. 3. Sugars and volatile fatty acid production after anaerobic digestion assays.

The top five dominant bacterial classes in the seed sludge were *Anaerolineae* (8.94 %), *Synergistia* (8.60 %), *Clostridia* (7.71 %), *Bacteroidia* (6.55 %) and *Gammaproteobacteria* (6.03 %); in the negative control were *Anaerolineae* (9.96 %), *Synergistia* (8.12 %), *Clostridia* (6.89 %), *Bacteroidia* (3.83 %) and *Syntrophobacteria* (3.64 %); in the SJ digestion were *Clostridia* (15.94 ± 1.39 %), *Anaerolineae* (9.53 ± 0.35 %), *Synergistia* (8.51 ± 0.72 %), *Bacteroidia* (4.05 ± 0.61 %) and *Gammaproteobacteria* (3.36 ± 1.05 %); and in the SP digestion were *Anaerolineae* (11.35 ± 1.14 %), *Clostridia* (9.36 ± 1.75 %), *Synergistia* (8.50 ± 0.83 %), *Bacteroidia* (5.45 ± 1.25 %), *Aminicenantia* (3.88 ± 0.49 %). For the archaeal sequences, the metataxonomic analysis could not infer most of the taxa below *phylum*. The most abundant archaeal taxon was the *phylum Halobacteriota*, which was also the predominant taxon in all samples. *Euryarchaeota* and *Thermoplasmatota* were also present, but with relative abundance <1 % in all samples. The relative abundances of *Halobacteriota* for each condition were 32.25 %, 41.81 %, 30.77 ± 2.35 % and 36.33 ± 5.24 % for the sludge seed, negative control, SJ and SP, respectively. It is noteworthy to mention that the SILVA database (v. 138) [22] named the *phylum* as *Halobacterota*, while the validly published according to the "List of Prokaryotic names with Standing in Nomenclature" (LPSN) [46] named *Halobacteriota*, which is also how it is addressed in "The Genome Taxonomy Database" (GTDB) [47].

The *Clostridia class*, the predominant bacterial group in SJ digestion, contains microorganisms capable of degrading cellulose, proteins, and various fatty acids, including long-chain fatty acids. They perform both acetogenesis and syntrophic acid degradation, producing butyric acid, butanol, ethanol, isopropanol, and acetone [48,49]. Bacteria of the *Clostridia* class have been reported to possess genes for the Wood-Ljungdahl pathway (WLP), which ferments sugars into organic acids. In this pathway, carbon dioxide is reduced to carbon monoxide and then converted to acetyl-CoA, using hydrogen as an electron donor. However, some microorganisms can perform the reverse pathway, known as Syntrophic Acetate Oxidation. In this process, they oxidize acetate into hydrogen and carbon dioxide while growing syntrophically with hydrogenotrophic methanogens, which use the hydrogen and carbon dioxide to produce CH_4 [50]. The predominance of this *class* in the SJ digestion might be correlated with the acids present in the

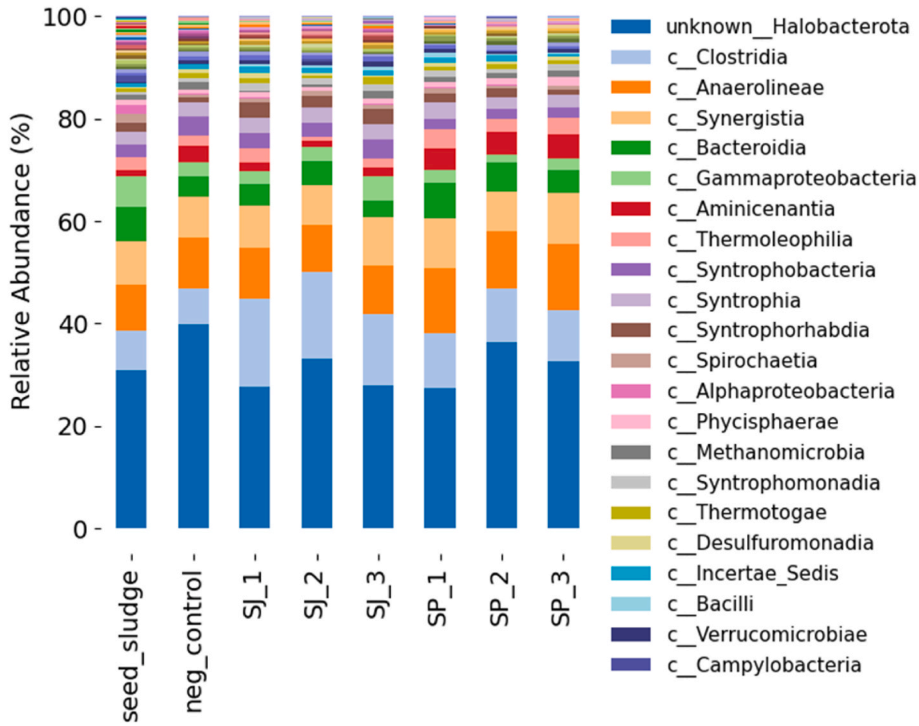

Fig. 4. Relative abundance of microorganisms in samples from each BMP test and seed sludge at class level.

substrate, syntrophically degrading and converting them into substrate for hydrogenotrophic archaea.

Anaerolineae is reported as one of core microbial populations in anaerobic digestion, being one of most present taxa in fermentations using pre-treated wastewater sludge [51] and sludge from mesophilic reactor fed with cattle dung [52]. *Anaerolineae* also produces hydrogen and carbon dioxide, and was reported possessing genes for carbohydrates hydrolases, including endoglucanases and beta-glycosidases, which could potentially make them capable of degrading cellulosic substrates [53]. Although, transcriptional analysis detected only a slightly cellulolytic activity [54]. Cai et al. (2021) [53] proposed that the addition of iron oxide enriches the *Anaerolineae* microorganisms and stimulates their carbohydrate and protein metabolism, increasing the CH_4 rate production in AD systems. Given that *Anaerolineae* is the most abundant bacterial group in SP digestion, their cellulolytic degradation pathways could be active. Additionally, the high amounts of Fe (Table 2) present in this substrate could be enhancing the microbial activity of this group.

The syntrophic bacteria of *Synergistia class* can be present in anaerobic digestion systems as consumers of complex organic materials and oxidizers of lactic and acetic acid to produce hydrogen and carbon dioxide [55]. These microorganisms act as exoelectrogens during direct interspecies electron transfer, providing energy for substrate degradation and indicating a syntrophic relationship with hydrogenotrophic methanogens [56]. The consistent abundance of *Synergistia* across SJ, SP, negative control and seed sludge samples shows that the agave substrates and process parameters did not affect this microbial population at the *class* level.

Bacteroidia also was reported as having some *genus* potentially involved in the hydrolysis and fermentation of proteins and polysaccharides during anaerobic digestion [57]. This *class* was associated with the acetate and propionate production, and along with *Clostridia*, were reported as predominant groups in anaerobic digestion of food waste [55]. Besides producers, *Bacteroidia* was also related with genes encoding for propionyl coenzyme A carboxylase, making these microorganisms potentially propionate oxidizers through the Methylmalonyl-CoA pathway [58], which might be contributing to the low levels of propionic acid observed in the digestion assays (Fig. 3).

Gammaproteobacteria was at the top five of most abundant *classes* in the seed sludge and in SJ digestion, but not in the negative control and in the SP digestion. In the SJ digestion, the abundance of these microorganisms was half of that of the seed sludge. This indicates a selective pressure of this *class* during the digestion assays, mainly in the process parameters conditions with a scarcity of substrates (negative control) and in more recalcitrant substrate (SP). Also, the competition with other classes might have occurred, since the *Gammaproteobacteria* functional roles is also the degradation of glucose and volatile fatty acids such as acetate, propionate and butyrate [59].

For the *Aminicentia* class, the only condition that had its abundance increased and placed at the top five was the SP digestion. Although being reported in AD experiments [56,60], no *Aminicentia* cultured representative exists, and its role in these systems is based on metagenome-assembled genomes (MAGs). MAGs analysis showed a potential for degradation and fermentation of complex carbohydrates and proteins, WLP driven acetate oxidation and carbon dioxide production through formate oxidation [61], acting as destructors of complex organic matter to produce hydrogen and acetate [62]. Considering the potential of multiple carbohydrate utilization, the presence of *Aminicentia* in the SP digestion might be due to its capacity to degrade the complex sugars in this substrate, mainly inulin and agavins commonly reported in agave [63].

The dominant archaeal group found in every sample was the *phylum Halobacteriota*. This *phylum* contains methanogenic microorganisms that can produce CH_4 through the aceticlastic, hydrogenotrophic and methylotrophic pathways, with the *Methanosarcina* and *Methanothrix* being the representative genera and well described in AD systems [64]. The high concentrations of K in agave substrate also could be a factor affecting the abundance of *Halobacteria*, since it is reported that K^+ increased the abundance of this *phylum* and allowed an efficient CH_4 production from anaerobic digestion of high salinity food waste [65]. The high abundance of *Halobacterota* with different methanogenesis pathways and the presence of syntrophic bacteria producing acetate, carbon dioxide and hydrogen, suggests that multiple pathways could be active simultaneously to produce CH_4 from agave substrates.

3.4. Alpha and beta-diversity of microbial community

To assess the effects of the AD of agave substrates in the diversity of each microbial community, the alpha-diversity was evaluated with the Shannon Entropy, presented in Fig. 5.

The Shannon Entropy did not differ between the seed sludge, negative control and SJ and SP digestion (Kruskal-Wallis, $p > 0.05$), meaning that, despite of the different conditions, all communities presented the similar alpha-diversity levels, i.e. the richness and evenness in each sample were the same. Although, the Shannon Entropy did not account for the phylogenetic relationship between the microorganisms in each sample, so to assess the differences between the communities (beta-diversity). Principal Coordinate Analysis (PCoA) for Unweighted Unifrac were used (Fig. 6). A clear separation between seed sludge, negative control, SJ and SP communities can be seen in the PCoA, with each microbial community clustering according to the experimental conditions.

The two different agave substrates (SJ and SP) were able to input a selective pressure that shaped the microbial communities, showing that the microorganism of each condition differs phylogenetically, indicating a specialization of the microbial communities to its respective substrate. Since that Unweighted Unifrac does not account for abundance, the difference between groups might be associated with the less abundant or lower-level taxa. These results suggest that despite the similarities in the abundance and classification of higher taxa, AD of agave occurs by molding the microbial communities in lower or less abundant taxa according to the characteristics of each substrate.

3.5. Energy biorefinery potential analysis

Based on the CH_4 production data obtained from batch experiments, energy analysis was performed for conversion into electrical and thermal energy within a CHP system, and in products from dry SP pyrolysis process: biochar and pyrolytic gas. This modification significantly boosts the overall energy yield, as demonstrated in Fig. 7.

As synthesized in Fig. 7, the total recoverable energy from the integrated system equals 1034 MJ for Scenario 1 (direct SP AD) and 1951 MJ for Scenario 2 (juice AD + pyrolysis configuration). These totals are computed as the sum of the contributions from anaerobic digestion credited within the study's system boundary and from pyrolysis

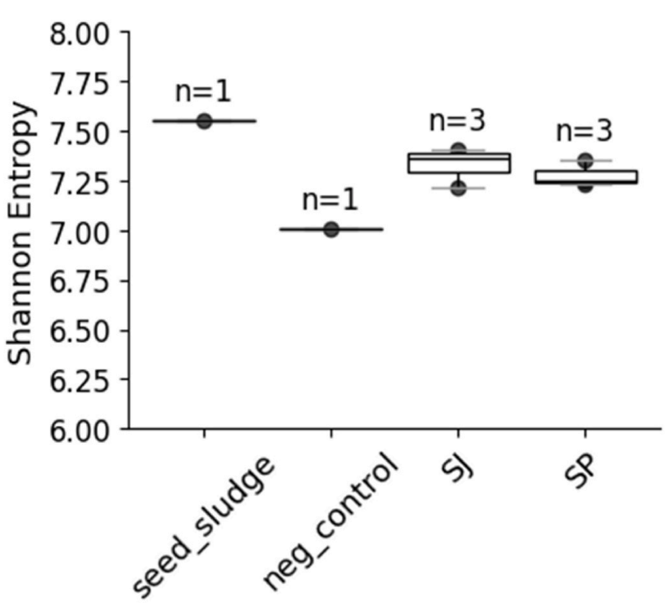

Fig. 5. Alpha-diversity analysis of microbial communities from SJ anaerobic digestion, SP anaerobic digestion, negative control and seed sludge. Kruskal-Wallis (all groups) $p > 0.05$.

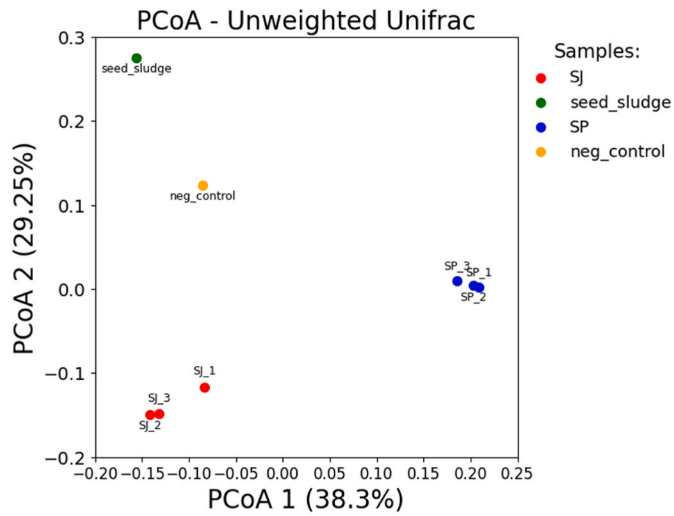

Fig. 6. Beta-diversity analysis of microbial communities from SJ anaerobic digestion, SP anaerobic digestion, negative control and seed sludge.

products. As defined in section 2.5, scenario 1, represented by the traced line, bioCH_4 is produced directly from the SP. In Scenario 2, SJ and dry SP are separated, with the dry SP then undergoing pyrolysis. This approach aligns with the simulation conducted by Díaz-Jiménez et al. (2019) [66]. The flowchart in Fig. 7 describes the energy generation potential of each product recovered for each scenario. The energy required for recovery of products (fiber, biochar, and bioCH_4) was taken into account to calculate the surplus energy and is detailed in Table 4, the unit operations are depicted by the lines connecting each scenario.

In general, thermal energy production is higher compared to electrical energy, which could be utilized for heating the AD system itself. The best scenario was for the SP residue when considering only bioCH_4 generation, even though the maximum volumetric CH_4 production was lower when compared to SJ (as described in Section 3.2).

Although the energy generation is higher for SP when focusing solely on bioCH_4 production, scenario with SJ and dry sisal pulp offers an additional advantage: two different products can be obtained through sisal fiber extraction, leading to a self-sufficient biorefinery with an electric energy surplus of 83.30 %. This surplus is due to the pyrolytic gas generated during biochar production, which can supply enough energy to sustain the entire sisal fiber production chain, including residue valorization.

In the case of scenario 2, biogas is not used to produce electricity. Instead, all the electrical energy required to obtain the substrates is supplied by the energy generated from the combustion of pyrolytic gas. In contrast, Scenario 1 proposes that 16 % of the energy produced is used for electricity generation, with 10 % of that allocated for biogas purification. Although the energetic gain is higher for SP, there are concerns about solid loading in anaerobic systems. This can lead to increased energy consumption during biomass preparation, if necessary, for example, some type of pre-treatment, which is commonly required for solid biomass, can be applied. As a result, dealing with solid wastes can lead to even higher energy expenditure as well as operational costs [42].

It is important to note that some solid-state bioreactors, which operate with low water content and minimal agitation, typically consume around 12 kJ $(m^3\ h)^{-1}$ of electrical energy for agitation [67]. In the present study, a low solids content of 5 % TS was used for SP, which likely contributed to the process's effective performance and higher methane yields. Conversely, using a bioreactor with higher solids content might have resulted in lower performance, requiring more energy and potentially leading to the accumulation of VFAs during operation [68]. Therefore, even though the SP condition demonstrated superior energy performance, a comprehensive evaluation for scaling up should

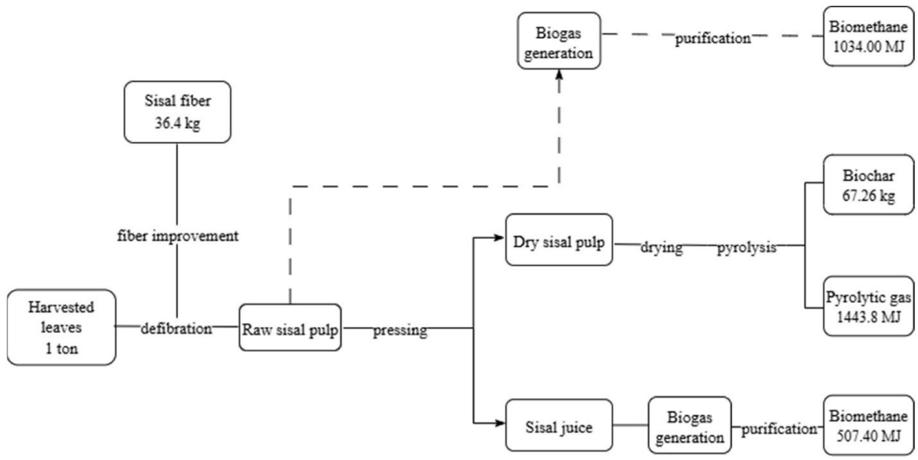

Fig. 7. Agave biorefinery flowchart: Utilization of sisal residues to energy generation through AD and pyrolysis.

Table 4
Energy consumption for each unit operation used to recover sisal fiber, biochar and bioCH_4.

Scenario 2		Scenario 1	
Electricity			
Defibration	8.90 %	Defibration	73.60 %
Press	4.70 %	Fiber improvement	26.40 %
Fiber improvement	3.20 %		
Surplus	83.30 %	Surplus	0 %
Heat			
Biogas purification	29.00 %	Biogas purification	59.40 %
Dry sisal pulp drying	3.50 %		
Pyrolysis	49.90 %		
Surplus	17.60 %	Surplus	40.60 %

consider the reactor type, solids content, and overall energy expenditures, including those for pumps and agitation [11]. This paper envisions a cooperative model that centralizes fiber processing and co-locates conversion units. Freshly pressed SJ is routed to anaerobic digestion to produce biogas for process heat, electricity (via CHP), while the pressed solid SP is dried with waste heat and sent to pyrolysis to generate additional energy carriers and biochar. Heat and mass integration improves overall efficiency, and the biochar + digestate can be returned to fields to enhance soil water retention and nutrient holding in semi-arid soils. This cooperative setup turns leaf processing residues into energy and soil-amendment products within a local, circular value chain.

4. Conclusions

SJ and SP residues are suitable for CH_4 production, demonstrating a high yield per gram of VS. Despite the high production, optimizing experiments to reduce water input for SJ and SP AD processes is necessary and is part of our ongoing research. SJ may be more suitable for CH_4 production due to its higher yield and no lag phase presentation, leading to greater production in less time. Additionally, working with a liquid substrate has advantages over a solid one, which can be used in other profitable processes. The microbial community composition varies based on the substrates, influencing the activity and abundance of the microorganisms. The community showed a saccharolytic/fermentative core dominated by *Clostridia* and *Bacteroidota*, supporting rapid hydrolysis/acidogenesis of SJ sugars. Syntrophic lineages were detected, coupling VFA oxidation via interspecies transfer. Methanogenesis was driven mainly by acetoclastic *Methanothrix* and *Methanosarcina*, with contributions from hydrogenotrophic taxa, consistent with the observed absent lag and high CH_4 yields. This functional consortium explains the performance reported and supports Scenario 2 as the most robust pathway under the tested regime.

Energy recovery suggests that combining AD of SJ with pyrolysis of dry sisal pulp is the best scenario for integrating agave biorefinery, and it can recover up to 1951 MJ of total energy. Agave-derived residues are particularly strategic in semi-arid regions because Agave sisalana is a drought-adapted CAM crop with exceptionally high water-use efficiency and low input requirements, thriving on marginal lands where conventional crops underperform. In this context, the configuration evaluated here: scenario 2 aligns with water-scarcity constraints: AD of the native liquid fraction (SJ) avoids dilution water, while pyrolyzing the solid fraction (SP) yields biochar that can improve soil water retention and nutrient holding in degraded semi-arid soils, closing material loops. Together with the higher total energy recovery reported here for Scenario 2, these attributes make the pathway well-suited for the Brazilian semi-arid belt.

This study highlights the suitability of A. *sisalana* residues for CH_4 production and shows how integrating them into a biorefinery linked to the fiber-extraction chain adds value while improving residue management. Accordingly, we recommend Scenario 2 as the most integrated and energy-efficient route under our conditions. Implemented via centralized fiber-processing hubs and cooperatives, it can mitigate environmental burdens from leaf-processing residues and diversify local renewable-energy supply.

CRediT authorship contribution statement

Fabiane M. Vieira: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Maiki S. de Paula:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis. **Maria Paula C. Volpi:** Writing – review & editing, Supervision, Formal analysis, Data curation, Conceptualization. **Oscar F. Herrera Adarme:** Writing – review & editing, Supervision, Methodology, Data curation, Conceptualization. **Thiago Ribas:** Writing – review & editing, Methodology, Investigation. **Jean C.G. Silva:** Writing – review & editing, Methodology, Investigation, Formal analysis. **Marcelo F. Carazzolle:** Writing – review & editing, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization. **Gonçalo A.G. Pereira:** Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Gustavo Mockaitis:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Acknowledgement

Agência Nacional do Petróleo, Gás Natural e Biocombustíveis do Brasil (ANP) in association with Shell Brasil Petróleo Ltda, with investments from the PD&I Clauses, for the Brazilian Agave Development Program (BRAVE) (research agreements 23018-5/22857-7).

Data availability

Data will be made available on request.

References

[1] C.E. Hollas, K.G.C. do Amaral, M.V. Lange, M.M. Higarashi, R.L. Radis Steinmetz, E.C. Barros, et al., Life cycle assessment of waste management from the Brazilian pig chain residues in two perspectives: electricity and biomethane production, J. Clean. Prod. 354 (2022) 131654, https://doi.org/10.1016/J.JCLEPRO.2022.131654.

[2] C.C. Anyaoku, S. Baroutian, Decentralized anaerobic digestion systems for increased utilization of biogas from municipal solid waste, Renew. Sustain. Energy Rev. 90 (2018) 982–991, https://doi.org/10.1016/J.RSER.2018.03.009.

[3] E.M.M. Esteves, A.M.N. Herrera, V.P.P. Esteves, C. do RV. Morgado, Life cycle assessment of manure biogas production: a review, J. Clean. Prod. 219 (2019) 411–423, https://doi.org/10.1016/J.JCLEPRO.2019.02.091.

[4] M.P.C. Volpi, A.D.N.F. Junior, T.T. Franco, B.S. Moraes, Operational and biochemical aspects of co-digestion (co-AD) from sugarcane vinasse, filter cake, and deacetylation liquor, Appl. Microbiol. Biotechnol. 105 (2021) 8969–8987, https://doi.org/10.1007/s00253-021-11635-x.

[5] M.L.M. Broeren, S.N.C. Dellaert, B. Cok, M.K. Patel, E. Worrell, L. Shen, Life cycle assessment of sisal fibre e Exploring how local practices can influence environmental performance. https://doi.org/10.1016/j.jclepro.2017.02.073, 2017.

[6] J. Arreola-Vargas, V. Ojeda-Castillo, R. Snell-Castro, R.I. Corona-González, F. Alatriste-Mondragón, H.O. Méndez-Acosta, Methane production from acid hydrolysates of Agave tequilana bagasse: evaluation of hydrolysis conditions and methane yield, Bioresour. Technol. 181 (2015) 191–199, https://doi.org/10.1016/J.BIORTECH.2015.01.036.

[7] J. Álvarez-Chávez, M. Villamiel, L. Santos-Zea, A.K. Ramírez-Jiménez, Agave By-products: an overview of their nutraceutical value, current applications, and processing methods, Polysaccharides 2 (2021) 720–743, https://doi.org/10.3390/POLYSACCHARIDES2030044.

[8] A. Mshandete, A. Kivaisi, M. Rubindamayugi, B. Mattiasson, Anaerobic batch co-digestion of sisal pulp and fish wastes, Bioresour. Technol. 95 (2004) 19–24, https://doi.org/10.1016/J.BIORTECH.2004.01.011.

[9] T.A. Colley, J. Valerian, M.Z. Hauschild, S.I. Olsen, M. Birkved, Addressing nutrient depletion in Tanzanian sisal fiber production using life cycle assessment and circular economy principles, with bioenergy Co-Production, Sustain. Times 13 (2021) 8881, https://doi.org/10.3390/SU13168881, 2021;13:8881.

[10] I. Angelidaki, M. Alves, D. Bolzonella, L. Borzacconi, J.L. Campos, A.J. Guwy, et al., Defining the biomethane potential (BMP) of solid organic wastes and energy crops: a proposed protocol for batch assays, Water Sci. Technol. 59 (2009) 927–934, https://doi.org/10.2166/WST.2009.040.

[11] H.K. Khuntia, A. Paliwal, D.R. Kumar, H.N. Chanakya, Review on solid-state anaerobic digestion of lignocellulosic biomass and organic solid waste, Environ. Monit. Assess. 194 (2022) 1–27, https://doi.org/10.1007/S10661-022-10160-2/FIGURES/2.

[12] B.E.L. Baêta, D.R.S. Lima, J.G.B. Filho, O.F.H. Adarme, L.V.A. Gurgel, SF de Aquino, Evaluation of hydrogen and methane production from sugarcane bagasse hemicellulose hydrolysates by two-stage anaerobic digestion process, Bioresour. Technol. 218 (2016) 436–446, https://doi.org/10.1016/J.BIORTECH.2016.06.113.

[13] American Public Health Association, American Water Works Association WEF, Standard Methods for the Examination of Water and Waste Water, 24th ed., 2023.

[14] S.H. Hosseini, X. Liang, C. Niyungeko, H. Miaomiao, F. Li, S. Khan, et al., Effect of sheep manure-derived biochar on colloidal phosphorus release in soils from various land uses, Environ. Sci. Pollut. Res. 26 (2019) 36367–36379, https://doi.org/10.1007/S11356-019-06762-Y/TABLES/6.

[15] P.R. Yaashikaa, P.S. Kumar, S. Varjani, A. Saravanan, A critical review on the biochar production techniques, characterization, stability and applications for circular bioeconomy, Biotechnol Reports 28 (2020) e00570, https://doi.org/10.1016/J.BTRE.2020.E00570.

[16] AG. de O. Paranhos, O.F.H. Adarme, G.F. Barreto, S. de Q. Silva, SF de Aquino, Methane production by co-digestion of poultry manure and lignocellulosic biomass: kinetic and energy assessment, Bioresour. Technol. 300 (2020) 122588, https://doi.org/10.1016/J.BIORTECH.2019.122588.

[17] F. Fantozzi, U. Desideri, R. De Prosperis, A. Russo, Gas turbines fired with biomass pyrolysis syngas: analysis of the overheating of hot gas path components, J. Eng. Gas Turbines Power 132 (2010) 1–8, https://doi.org/10.1115/1.4000134/465437.

[18] R. Cano, S.I. Pérez-Elvira, F. Fdz-Polanco, Energy feasibility study of sludge pretreatments: a review, Appl. Energy 149 (2015) 176–185, https://doi.org/10.1016/J.APENERGY.2015.03.132.

[19] S. Chen, Y. Zhou, Y. Chen, J. Gu, Fastp: an ultra-fast all-in-one FASTQ preprocessor, Bioinformatics 34 (2018) i884–i890, https://doi.org/10.1093/BIOINFORMATICS/BTY560.

[20] E. Bolyen, J.R. Rideout, M.R. Dillon, N.A. Bokulich, C.C. Abnet, G.A. Al-Ghalith, et al., Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2, Nat. Biotechnol. 37 (8) (2019) 852–857, https://doi.org/10.1038/s41587-019-0209-9, 2019;37.

[21] B.J. Callahan, P.J. McMurdie, M.J. Rosen, A.W. Han, A.J.A. Johnson, S.P. Holmes, DADA2: high-resolution sample inference from Illumina amplicon data, Nat. Methods 13 (7) (2016) 581–583, https://doi.org/10.1038/nmeth.3869, 2016;13.

[22] C. Quast, E. Pruesse, P. Yilmaz, J. Gerken, T. Schweer, P. Yarza, et al., The SILVA ribosomal RNA gene database project: improved data processing and web-based tools, Nucleic Acids Res. 41 (2013) D590–D596, https://doi.org/10.1093/NAR/GKS1219.

[23] C.E. Shannon, A mathematical theory of communication, Bell Syst Tech J 27 (1948) 379–423, https://doi.org/10.1002/J.1538-7305.1948.TB01338.X.

[24] C. Lozupone, M.E. Lladser, D. Knights, J. Stombaugh, R. Knight, UniFrac: an effective distance metric for microbial community comparison, ISME J. 5 (2) (2011) 169–172, https://doi.org/10.1038/ismej.2010.133, 2010;5.

[25] B.S. Moraes, M. Zaiat, A. Bonomi, Anaerobic digestion of vinasse from sugarcane ethanol production in Brazil: challenges and perspectives, Renew. Sustain. Energy Rev. 44 (2015) 888–903, https://doi.org/10.1016/j.rser.2015.01.023.

[26] V.L. Soares, M. de O. Faber, A.F. Monteiro, M.C. Cammarota, V.S. Ferreira-Leitão, Potential use of sisal juice as raw material for sequential biological production of hydrogen and methane, Brazilian J Chem Eng (2023), https://doi.org/10.1007/S43153-023-00342-X.

[27] B. Dridi, M.L. Fardeau, B. Ollivier, D. Raoult, M. Drancourt, Methanomassiliicoccus luminyensis gen. nov., sp. Nov., a methanogenic archaeon isolated from human faeces, Int. J. Syst. Evol. Microbiol. 62 (2012) 1902–1907, https://doi.org/10.1099/IJS.0.033712-0/CITE/REFWORKS.

[28] C. Mao, Y. Feng, X. Wang, G. Ren, Review on research achievements of biogas from anaerobic digestion, Renew. Sustain. Energy Rev. 45 (2015) 540–555, https://doi.org/10.1016/J.RSER.2015.02.032.

[29] T. Zhang, P. Zhang, Z. Hu, Q. Qi, Y. He, J. Zhang, New insight on Fe-bioavailability: Bio-uptake, utilization and induce in optimizing methane production in anaerobic digestion, Chem Eng J 441 (2022) 136099, https://doi.org/10.1016/J.CEJ.2022.136099.

[30] Y. Zhang, Z. Zhang, K. Suzuki, T. Maekawa, Uptake and mass balance of trace metals for methane producing bacteria, Biomass Bioenergy 25 (2003) 427–433, https://doi.org/10.1016/S0961-9534(03)00012-6.

[31] O. Berzal de Frutos, M. Götze, M. Pidou, Y. Bajón Fernández, Anaerobic Co-Digestion of sewage sludge and trade wastes: beneficial and inhibitory effects of individual constituents, Processes 11 (2023) 519, https://doi.org/10.3390/PR11020519/S1.

[32] J.H. Ahn, T.H. Do, S.D. Kim, S. Hwang, The effect of calcium on the anaerobic digestion treating swine wastewater, Biochem. Eng. J. 30 (2006) 33–38, https://doi.org/10.1016/J.BEJ.2006.01.014.

[33] J. Zhao, Y. Li, S. Pan, Q. Tu, W. Dang, Z. Wang, et al., Effects of magnesium chloride on the anaerobic digestion and the implication on forward osmosis membrane bioreactor for sludge anaerobic digestion, Bioresour. Technol. 268 (2018) 700–707, https://doi.org/10.1016/J.BIORTECH.2018.08.002.

[34] L. Appels, J. Baeyens, J. Degrè Ve, R. Dewil, Principles and potential of the anaerobic digestion of waste-activated sludge ARTICLE IN PRESS, Prog. Energy Combust. Sci. 34 (2008) 755–781, https://doi.org/10.1016/j.pecs.2008.06.002.

[35] H. Jung, J. Kim, C. Lee, Temperature effects on methanogenesis and sulfidogenesis during anaerobic digestion of sulfur-rich macroalgal biomass in sequencing batch reactors, Microorg 7 (2019) 682, https://doi.org/10.3390/MICROORGANISMS7120682, 2019;7:682.

[36] M. Zhang, Z. Fan, Z. Hu, X. Luo, Enhanced anaerobic digestion with the addition of chelator-nickel complexes to improve nickel bioavailability, Sci. Total Environ. 759 (2021) 143458, https://doi.org/10.1016/J.SCITOTENV.2020.143458.

[37] O. Cundr, D. Haladova, Biogas Yield from Anaerobic Batch Co-Digestion of Sisal Pulp and Zebu Dung, 2015.

[38] K. Lueangwattanapong, F. Ammam, P.M. Mason, C. Whitehead, S.J. McQueen-Mason, L.D. Gomez, et al., Anaerobic digestion of Crassulacean Acid Metabolism plants: exploring alternative feedstocks for semi-arid lands, Bioresour. Technol. 297 (2020) 122262, https://doi.org/10.1016/j.biortech.2019.122262.

[39] F. Rizzioli, C. Magonara, G. Mengoli, D. Bolzonella, F. Battista, Production, purification and recovery of caproic acid, volatile fatty acids and methane from Opuntia ficus indica, Renew. Sustain. Energy Rev. 190 (2024) 114083, https://doi.org/10.1016/j.rser.2023.114083.

[40] M. Barragán-Trinidad, G. Buitrón, Pretreatment of agave bagasse with ruminal fluid to improve methane recovery, Waste Manag. 175 (2024) 52–61, https://doi.org/10.1016/J.WASMAN.2023.12.041.

[41] I. Valdez-Vazquez, F. Alatriste-Mondragón, J. Arreola-Vargas, G. Buitrón, J. Carillo-Reyes, E. León-Becerril, et al., A comparison of biological, enzymatic, chemical and hydrothermal pretreatments for producing biomethane from Agave bagasse, Ind. Crops Prod. 145 (2020) 112160, https://doi.org/10.1016/J.INDCROP.2020.112160.

[42] I. Ortíz, I. Valdez-Vazquez, A. Hernández-Vázquez, R. Olivares-Hernández, J. Carillo-Reyes, F. Alatriste-Mondragón, et al., Process design and economic evaluation for methane and electricity production from Agave bagasse using different pretreatments, Energy Convers. Manag. 306 (2024) 118319, https://doi.org/10.1016/J.ENCONMAN.2024.118319.

[43] De Paula MP, Lacerda TM, Zambon MD, Frollini E. Adding value to the Brazilian sisal: acid hydrolysis of its pulp seeking production of sugars and materials n.d. https://doi.org/10.1007/s10570-012-9674-8.
[44] M. Yadav, C. Joshi, K. Paritosh, J. Thakur, N. Pareek, S.K. Masakapalli, et al., Organic waste conversion through anaerobic digestion: a critical insight into the metabolic pathways and microbial interactions, Metab. Eng. 69 (2022) 323–337, https://doi.org/10.1016/J.YMBEN.2021.11.014.
[45] A. Rahimieh, M. Nosrati, A review on biochemistry, microbiology and thermodynamic aspects of propionate: the key intermediate in the anaerobic digestion and wastewater treatment, Desalin Water Treat 317 (2024) 100191, https://doi.org/10.1016/J.DWT.2024.100191.
[46] A.C. Parte, J.S. Carbasse, J.P. Meier-Kolthoff, L.C. Reimer, M. Göker, List of prokaryotic names with Standing in Nomenclature (LPSN) moves to the DSMZ, Int. J. Syst. Evol. Microbiol. 70 (2020) 5607–5612, https://doi.org/10.1099/IJSEM.0.004332.
[47] D.H. Parks, M. Chuvochina, C. Rinke, A.J. Mussig, P.A. Chaumeil, P. Hugenholtz, GTDB: an ongoing census of bacterial and archaeal diversity through a phylogenetically consistent, rank normalized and complete genome-based taxonomy, Nucleic Acids Res. 50 (2022) D785–D794, https://doi.org/10.1093/NAR/GKAB776.
[48] C. Cho, Y.S. Jang, H.G. Moon, J. Lee, S.Y. Lee, Metabolic engineering of clostridia for the production of chemicals, Biofuels, Bioprod Biorefining 9 (2015) 211–225, https://doi.org/10.1002/BBB.1531.
[49] A. Schnürer, Biogas production: microbiology and technology, Adv. Biochem. Eng. Biotechnol. 156 (2016) 195–234, https://doi.org/10.1007/10_2016_5.
[50] S. Campanaro, L. Treu, P.G. Kougias, D. De Francisci, G. Valle, I. Angelidaki, Metagenomic analysis and functional characterization of the biogas microbiome using high throughput shotgun sequencing and a novel binning strategy, Biotechnol. Biofuels 9 (2016) 1–17, https://doi.org/10.1186/S13068-016-0441-1/FIGURES/5.
[51] H. Ma, H. Liu, L. Zhang, M. Yang, B. Fu, H. Liu, Novel insight into the relationship between organic substrate composition and volatile fatty acids distribution in acidogenic co-fermentation, Biotechnol. Biofuels 10 (2017) 1–15, https://doi.org/10.1186/S13068-017-0821-1/FIGURES/7.
[52] X. Zhou, Y. Lu, L. Huang, Q. Zhang, X. Wang, J. Zhu, Effect of pH on volatile fatty acid production and the microbial community during anaerobic digestion of Chinese cabbage waste, Bioresour. Technol. 336 (2021) 125338, https://doi.org/10.1016/J.BIORTECH.2021.125338.
[53] C. Cai, L. Li, Y. Hua, H. Liu, X. Dai, Ferroferric oxide promotes metabolism in Anaerolineae other than microbial syntrophy in anaerobic methanogenesis of antibiotic fermentation residue, Sci. Total Environ. 758 (2021), https://doi.org/10.1016/J.SCITOTENV.2020.143601.
[54] Y. Xia, Y. Wang, Y. Wang, F.Y.L. Chin, T. Zhang, Cellular adhesiveness and cellulolytic capacity in Anaerolineae revealed by omics-based genome interpretation, Biotechnol. Biofuels 9 (2016), https://doi.org/10.1186/S13068-016-0524-Z.
[55] L. Li, Q. He, Y. Ma, X. Wang, X. Peng, A mesophilic anaerobic digester for treating food waste: process stability and microbial community analysis using pyrosequencing, Microb. Cell Fact. 15 (2016), https://doi.org/10.1186/S12934-016-0466-Y.
[56] Y. Li, W. Kong, H. Liu, Y. Hong, T. Huang, Enhanced degradation of phenolic compounds in coal gasification wastewater by activated carbon-Fe3O4 nanoparticles coupled with anaerobic co-metabolism, Biochem. Eng. J. 189 (2022) 108717, https://doi.org/10.1016/J.BEJ.2022.108717.
[57] A. Parsy, E. Ficara, V. Mezzanotte, A. Guerreschi, R. Guyoneaud, F. Monlau, et al., Incorporating saline microalgae biomass in anaerobic digester treating sewage sludge: impact on performance and microbial populations, Bioresour. Technol. 397 (2024), https://doi.org/10.1016/J.BIORTECH.2024.130444.
[58] J. Xu, L. Xu, Y. Zong, R. Lin, Y. He, L. Xie, Electrically-assisted anaerobic digestion under ammonia stress: facilitating propionate oxidation and activating methanogenesis via direct interspecies electron transfer, Bioresour. Technol. 393 (2024), https://doi.org/10.1016/J.BIORTECH.2023.130067.
[59] A.M. Hailu, S.G. Palani, S.L. Asfaw, T.A. Tegaye, Insight into microbial community diversity and composition of two-stage anaerobic digestion: focusing methanogenic stage, Bioresour Technol Reports 15 (2021) 100764, https://doi.org/10.1016/J.BITEB.2021.100764.
[60] Y. Xiang, W. Xiong, Z. Yang, R. Xu, Y. Zhang, M. Wu, et al., Metagenomic insights into the toxicity of carbamazepine to functional microorganisms in sludge anaerobic digestion, Sci. Total Environ. 919 (2024) 170780, https://doi.org/10.1016/J.SCITOTENV.2024.170780.
[61] A.E. Booker, T. D'Angelo, A. Adams-Beyea, J.M. Brown, O. Nigro, M.S. Rappé, et al., Life strategies for Aminicenantia in subseafloor oceanic crust, ISME J. 17 (9) (2023) 1406–1415, https://doi.org/10.1038/s41396-023-01454-5, 2023;17.
[62] V.V. Kadnikov, A.V. Mardanov, A.V. Beletsky, O.V. Karnachuk, N.V. Ravin, Genome of the candidate phylum Aminicenantes bacterium from a deep subsurface thermal aquifer revealed its fermentative saccharolytic lifestyle, Extremophiles 23 (2019) 189–200, https://doi.org/10.1007/S00792-018-01073-5/FIGURES/3.
[63] M.G. López, L.F. Salomé-Abarca, The agavins (Agave carbohydrates) story, Carbohydr. Polym. 327 (2024) 121671, https://doi.org/10.1016/J.CARBPOL.2023.121671.
[64] S.L. Bräuer, N. Basiliko, H.M.P. Siljanen, S.H. Zinder, Methanogenic archaea in peatlands, FEMS Microbiol. Lett. 367 (2020), https://doi.org/10.1093/FEMSLE/FNAA172.
[65] Y. Wang, J. Li, M. Liu, L. Gu, L. Xu, J. Li, et al., Enhancement of anaerobic digestion of high salinity food waste by magnetite and potassium ions: digestor performance, microbial and metabolomic analyses, Bioresour. Technol. 388 (2023) 129769, https://doi.org/10.1016/J.BIORTECH.2023.129769.
[66] L. Díaz-Jiménez, S. Carlos-Hernandez, D. Jasso de Rodríguez, R. Rodríguez-García, Conceptualization of a biorefinery for guishe revalorization, Ind. Crops Prod. 138 (2019) 111441, https://doi.org/10.1016/J.INDCROP.2019.06.004.
[67] J. Lu, H.N. Gavala, I.V. Skiadas, Z. Mladenovska, B.K. Ahring, Improving anaerobic sewage sludge digestion by implementation of a hyper-thermophilic prehydrolysis step, J Environ Manage 88 (2008) 881–889, https://doi.org/10.1016/J.JENVMAN.2007.04.020.
[68] W. Li, R. Gupta, Z. Zhang, L. Cao, Y. Li, P.L. Show, et al., A review of high-solid anaerobic digestion (HSAD): from transport phenomena to process design, Renew. Sustain. Energy Rev. 180 (2023) 113305, https://doi.org/10.1016/J.RSER.2023.113305.